CHAPTER 11

Measurement of variables: Operational definition

LEARNING OBJECTIVES

After completing Chapter 11, you should be able to:

1. Explain how variables are measured.
2. Explain when operationalization of variables is necessary.
3. Operationally define (or operationalize) abstract and subjective variables.

INTRODUCTION

Measurement of the variables is an integral part of research and an important aspect of research design (see shaded portion in Figure 11.1). Unless the variables are measured in some way, we will not be able to find answers to our research questions. Surveys and experimental designs, discussed in the previous chapters, often use questionnaires to measure the variables of interest. In this chapter we will discuss how variables lend themselves to measurement.

HOW VARIABLES ARE MEASURED

To test the hypothesis that workforce diversity affects organizational effectiveness we have to measure workforce diversity and organizational effectiveness. *Measurement* is the assignment of numbers or other symbols to characteristics (or attributes) of objects according to a prespecified set of rules. *Objects* include persons, strategic business units, companies, countries, bicycles, elephants, kitchen appliances, restaurants, shampoo, yogurt, and so on. Examples of *characteristics* of objects are arousal-seeking tendency, achievement motivation, organizational effectiveness, shopping enjoyment, length, weight, ethnic diversity, service quality, conditioning effects, and taste. It is important that you realize that you cannot measure objects (for instance, a company); you measure characteristics or attributes of objects (for instance, the organizational effectiveness of a company). In a similar fashion, you can